

| $\mathcal{K}_{0^+}^{\#1} \dagger$ | $\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$ | $\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$      | $\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$      | $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$ |
|-----------------------------------|-----------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-----------------------------------------------------|
| 0                                 | 0                                                   | $-\frac{Y_1 k^2}{5}$                            | $-\frac{Y_1 k^2}{5\sqrt{2}}$                    | 0                                               | 0                                               | 0                                               | 0                                                   |
| 0                                 | 0                                                   | $-\frac{Y_1 k^2}{5\sqrt{2}}$                    | $-\frac{Y_1 k^2}{10}$                           | 0                                               | 0                                               | 0                                               | $-\frac{9k^2}{10} \frac{{}^{(4)}k_1}{k_1}$          |
|                                   |                                                     | 0                                               | 0                                               | $\frac{k^2}{10} \frac{{}^{(4)}k_1}{k_1}$        | $\frac{k^2}{5\sqrt{2}} \frac{{}^{(4)}k_1}{k_1}$ |                                                 |                                                     |
|                                   |                                                     | 0                                               | 0                                               | $\frac{k^2}{5\sqrt{2}} \frac{{}^{(4)}k_1}{k_1}$ | $\frac{k^2}{5} \frac{{}^{(4)}k_1}{k_1}$         |                                                 |                                                     |

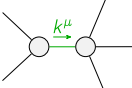
| $\mathcal{T}_{0^+}^{\#1}$                            | $\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                     |                                                             |                                                             |                                                      |                                               |
|------------------------------------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|
| $\mathcal{T}_{0^+}^{\#1\dagger}$                     | 0                                             | 0                                   |                                                             |                                                             |                                                      |                                               |
| $\mathcal{T}_{0^+}^{\#1\dagger}{}^{\alpha\beta\chi}$ | 0                                             | 0                                   | $\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$                   | $\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$                   | $\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$                 | $\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$          |
| $\mathcal{T}_{1^+}^{\#1\dagger}{}^{\alpha\beta}$     | $\frac{2}{3\text{Det}(1^+)}$                  | $\frac{\sqrt{2}}{3\text{Det}(1^+)}$ | 0                                                           | 0                                                           |                                                      |                                               |
| $\mathcal{T}_{1^+}^{\#2\dagger}{}^{\alpha\beta}$     | $\frac{\sqrt{2}}{3\text{Det}(1^+)}$           | $\frac{1}{3\text{Det}(1^+)}$        | 0                                                           | 0                                                           |                                                      |                                               |
| $\mathcal{T}_{1^+}^{\#1\dagger}{}^{\alpha}$          | 0                                             | 0                                   | $\frac{10}{9k^2} \frac{{}^{(4)}\kappa_1}{\kappa_1}$         | $\frac{10\sqrt{2}}{9k^2} \frac{{}^{(4)}\kappa_1}{\kappa_1}$ |                                                      |                                               |
| $\mathcal{T}_{1^+}^{\#2\dagger}{}^{\alpha}$          | 0                                             | 0                                   | $\frac{10\sqrt{2}}{9k^2} \frac{{}^{(4)}\kappa_1}{\kappa_1}$ | $\frac{20}{9k^2} \frac{{}^{(4)}\kappa_1}{\kappa_1}$         | $\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$            | $\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |
|                                                      |                                               |                                     | $\mathcal{T}_{2^+}^{\#1\dagger}{}^{\alpha\beta}$            | 0                                                           | 0                                                    |                                               |
|                                                      |                                               |                                     | $\mathcal{T}_{2^+}^{\#1\dagger}{}^{\alpha\beta\chi}$        | 0                                                           | $-\frac{10}{9k^2} \frac{{}^{(4)}\kappa_1}{\kappa_1}$ |                                               |

Abbreviations used in matrices

$$Y_1 = 6 \frac{{}^{(4)}\kappa_1}{\kappa_1} + 5 \frac{{}^{(4)}\kappa_3}{\kappa_3} \text{ \& Det}(1^+) = -\frac{3}{10} k^2 (6 \frac{{}^{(4)}\kappa_1}{\kappa_1} + 5 \frac{{}^{(4)}\kappa_3}{\kappa_3})$$

Lagrangian

$$\begin{aligned} & \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_{\alpha}^{\alpha\beta} - \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_{\alpha}^{\alpha\beta} + \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta - \\ & \frac{18}{5} \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{9}{5} \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \\ & \frac{12}{5} \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial^\chi \mathcal{K}_{\alpha}^{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \frac{1}{2} \frac{{}^{(4)}\kappa_3}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{3}{5} \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - \frac{3}{5} \frac{{}^{(4)}\kappa_1}{\kappa_1} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

|                                                                                    |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                        |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                        |
| Source constraint(s)                                                               | # constraint(s)                                                                      | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                        |
| $\mathcal{J}_{0^+}^{\#1 \alpha\beta\chi} == 0$                                     | 1                                                                                    | $\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$<br>$\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$ |                                                                                        |
| $\mathcal{J}_{0^+}^{\#1} == 0$                                                     | 1                                                                                    | $\partial_\beta \mathcal{J}_{\alpha}^{\alpha\beta} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                        |
| $\mathcal{J}_{1^+}^{\#1 \alpha} == \mathcal{J}_{1^+}^{\#2 \alpha}$                 | 3                                                                                    | $\partial_\chi \partial^\alpha \mathcal{J}_{\beta}^{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}_{\beta}^{\beta\alpha} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                        |
| $\mathcal{J}_{1^+}^{\#1 \alpha\beta} == \mathcal{J}_{1^+}^{\#2 \alpha\beta}$       | 3                                                                                    | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                        |
| $\mathcal{J}_{2^+}^{\#1 \alpha\beta} == 0$                                         | 5                                                                                    | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi\delta} == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_{\chi}^{\chi\delta} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                               |                                                                                        |
| Total # constraint(s):                                                             | 13                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                        |
| Resolved pole(s)                                                                   | # polarization(s)                                                                    | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Residue                                                                                |
|  | 2                                                                                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\frac{1}{-6 \frac{{}^{(4)}\kappa_1}{\kappa_1} - 5 \frac{{}^{(4)}\kappa_3}{\kappa_3}}$ |
| Resolved unitarity condition(s):                                                   | $\frac{{}^{(4)}\kappa_1}{\kappa_1} < -\frac{5}{6} \frac{{}^{(4)}\kappa_3}{\kappa_3}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                        |